Solution

Run 5

2110201: COMP ENG MATH (2023/2) Quiz 6

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1. (4 points) Find an orthonormal basis that spans the same subspace as $\left\langle \overset{u_1}{\begin{bmatrix}1\\1\\2\end{bmatrix}}, \overset{u_2}{\begin{bmatrix}0\\6\\9\end{bmatrix}}, \overset{u_3}{\begin{bmatrix}4\\2\\0\end{bmatrix}} \right\rangle$

by using **Gram-Schmidt algorithm** only.

Let $v_1 = u_1 = \begin{bmatrix}1\\1\\2\end{bmatrix} \xrightarrow{\text{normalize}} \begin{bmatrix}1/\sqrt{6}\\1/\sqrt{6}\\2/\sqrt{6}\end{bmatrix}$

$v_2 = u_2 - \frac{u_2 \cdot v_1}{v_1 \cdot v_1}\begin{bmatrix}1\\1\\2\end{bmatrix} = \begin{bmatrix}0\\6\\9\end{bmatrix} - 4\begin{bmatrix}1\\1\\2\end{bmatrix} = \begin{bmatrix}-4\\2\\1\end{bmatrix} \xrightarrow{\text{normalize}} \begin{bmatrix}-4/\sqrt{21}\\2/\sqrt{21}\\1/\sqrt{21}\end{bmatrix}$

$v_3 = u_3 - \frac{u_3 \cdot v_1}{v_1 \cdot v_1}\begin{bmatrix}1\\1\\2\end{bmatrix} - \frac{u_3 \cdot v_2}{v_2 \cdot v_2}\begin{bmatrix}-4\\2\\1\end{bmatrix} = \begin{bmatrix}4\\2\\0\end{bmatrix} - 1\begin{bmatrix}1\\1\\2\end{bmatrix} + \frac{4}{7}\begin{bmatrix}-4\\2\\1\end{bmatrix} = \begin{bmatrix}5/7\\15/7\\-10/7\end{bmatrix} \xrightarrow[\left(\begin{bmatrix}1\\3\\-2\end{bmatrix}\right)]{\text{normalize}} \begin{bmatrix}1/\sqrt{14}\\3/\sqrt{14}\\-2/\sqrt{14}\end{bmatrix}$

$\therefore$ The answer is $\langle v_1, v_2, v_3 \rangle = \left\langle \begin{bmatrix}1/\sqrt{6}\\1/\sqrt{6}\\2/\sqrt{6}\end{bmatrix}, \begin{bmatrix}-4/\sqrt{21}\\2/\sqrt{21}\\1/\sqrt{21}\end{bmatrix}, \begin{bmatrix}1/\sqrt{14}\\3/\sqrt{14}\\-2/\sqrt{14}\end{bmatrix} \right\rangle$

2. (4 points, 1 point each) Given a basis $B = \left\langle \begin{bmatrix}0\\0\\1/2\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}1/3\\0\\0\end{bmatrix} \right\rangle$

2.1. The matrix $\begin{bmatrix}0&0&2\\0&1&0\\3&0&0\end{bmatrix}$ maps vectors from the standard basis to vectors in the basis B.

2.2. The matrix $\begin{bmatrix}0&0&1/3\\0&1&0\\1/2&0&0\end{bmatrix}$ maps vectors from the basis B to vectors in the standard basis.

2.3. A vector $\begin{bmatrix}-1\\1\\-2\end{bmatrix}$ in a standard basis is a vector $\begin{bmatrix}-4\\1\\-3\end{bmatrix}$ in the basis B.

2.4. Given a basis $\mathrm{D} = \left\langle \begin{bmatrix}-2\\1\end{bmatrix}, \begin{bmatrix}1\\-1\end{bmatrix} \right\rangle$ and a transformation T represented by a matrix $\mathbf{A} = \begin{bmatrix}2&-1\\3&0\\-1&5\end{bmatrix}$, write $\mathrm{Rep}_{D,B}(T)$ as a multiplication of 3 matrices. (No need to compute the actual value.)

$$B^{-1}AD = \begin{bmatrix}0&0&2\\0&1&0\\3&0&0\end{bmatrix}\begin{bmatrix}2&-1\\3&0\\-1&5\end{bmatrix}\begin{bmatrix}-2&1\\1&-1\end{bmatrix} \quad \#$$

3. (2 points, 1 point each) Answer the following questions with a briefly explanation.

3.1. Given a transformation S: $\mathbf{R}^2 \Rightarrow \mathbf{R}^2$, given by $S(x, y) = \frac{1}{2}(x + y, x + y)$. Is S distortion-free?

We need to check if $S(x_1, y_1) \cdot S(x_2, y_2) = (x_1, y_1) \cdot (x_2, y_2)$?

$\frac{1}{2}(x_1+y_1, x_1+y_1) \cdot \frac{1}{2}(x_2+y_2, x_2+y_2) = x_1x_2 + y_1y_2$?

$\frac{1}{4}\left[2(x_1+y_1)(x_2+y_2)\right] = x_1x_2 + y_1y_2$? False #

3.2. What is **linear interpolation**? Can it be described using knowledges from **the change of basis**?

↳ I give free on this, answer whatever you feel like is the best definition.

the matrix or the transformation we want to find.

$$\begin{tikzcd} R_n \arrow[r, "A"] & R_n \\ \langle B \rangle \arrow[u, "B"] \arrow[ur, "W"'] & \end{tikzcd}$$

change of basis from B to R_n

↳ given transformation to a basis.

Drawing diagrams or explaination like this would give you full credit.

$A = WB^{-1}$

(or $W = \mathrm{Rep}_{B,I_n}(A)$)